

Forecasting Marine Heatwaves in Alaska Shelf Seas: Methods, Predictability, and Validation Challenges

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Abstract. Despite rapid progress in global and regional marine-heatwave (MHW) forecasting, there is little direct evidence on forecast skill for the Gulf of Alaska and the Bering, Chukchi, and Beaufort shelves. The methodological approaches available for such forecasts — coupled dynamical seasonal systems, machine-learning and hybrid models, linear inverse models, and statistical baselines — are mature in open-ocean, temperate, and tropical systems, and the principal physical mechanisms governing high-latitude sea-surface temperature (SST) persistence have been extensively investigated. The principal gap is the absence of region-specific evaluation. The central scientific problem is not the absence of forecasting methods; it is the absence of validated, benchmarked, trend-aware, probabilistic forecast evaluation for Alaska shelf systems. This synthesis is organized around the distinction between forecasting methodology, forecast validation, and physical predictability: it distinguishes existing forecast methodologies from the evidence on their forecast skill, the mechanisms that may confer predictability, and the outstanding methodological challenges specific to Alaska shelf seas, and closes with the elements of a credible regional evaluation design. The focus is the North Pacific high latitudes; sub-Arctic Atlantic systems are considered only for comparison, and the distinct physical regimes of the deeper, open-ocean-influenced Gulf of Alaska and the shallow, seasonally ice-affected Bering and Arctic shelves are treated as separate problems.

1. The predictand and the evaluation problem

Forecast evaluation begins with the definition of the predictand (the quantity to be predicted), and the relevant studies do not share a common predictand. Five distinct targets appear in the literature and should not be conflated: (i) SST anomaly forecasting, a continuous prediction; (ii) MHW binary occurrence, a threshold-exceedance event; (iii) MHW intensity and duration, including onset, decline, and cumulative magnitude; (iv) probabilistic event forecasting, the calibrated probability of an MHW; and (v) shelf and bottom thermal-state forecasting, addressing subsurface or near-bottom temperature as distinct from the surface. These predictands are not interchangeable. A method validated for one does not automatically transfer to another, and calibrated probabilistic occurrence skill in particular is not implied by a low SST-anomaly root-mean-square error.

The evaluation criterion matters as much as the predictand. For a climate-econometrics readership the relevant question is whether a model demonstrates superior out-of-sample probabilistic skill relative to persistence and climatology, under a transparent treatment of trend and regime change. Persistence and climatology are therefore not minor baselines but the central benchmarks; a model that does not improve upon persistence is of limited operational value. This standard is exacting at high latitudes precisely because oceanic thermal inertia makes persistence strong: at subsurface and high-heat-content scales, simple persistence frequently outperforms dynamical models (Smith &

Spillman 2024; McAdam et al. 2023), and damped-persistence and persistence-transformation schemes are competitive benchmarks in their own right (Lee & Tzeng 2012; Ross & Stock 2022). Any claim of skill for an Alaska shelf system must be stated relative to these benchmarks and under an explicit treatment of the warming trend (Section 5).

2. Existing forecast methodologies

The methodological repertoire is well developed and spans five families, each typically oriented toward a particular predictand.

- **Coupled dynamical seasonal systems** — NMME (Jacox et al. 2022), ECMWF SEAS5 (de Boissésou & Balmaseda 2024; Liu et al. 2025), ACCESS-S2 (Spillman et al. 2021; Smith & Spillman 2024), CMCC-SPS3.5 (McAdam et al. 2023), and NUIST-CFS1.0 (Zhang et al. 2023; Tang et al. 2024) are the principal operational approach at lead times of one to three months, primarily for MHW occurrence and probabilistic event forecasts. Their documented predictive skill is robust but strongly modulated by ENSO phase and concentrated outside the sub-Arctic.
- **Machine-learning and deep-learning methods** — long short-term memory networks, random forests, and convolutional architectures (Bonino et al. 2024; Giamalaki et al. 2022), U-Net models (Taylor & Feng 2022; Parasyris et al. 2025; Howard et al. 2026), and the MHWUNet model (Sun et al. 2024) — target SST anomalies and event occurrence and perform best at lead times of one day to two weeks; the available evidence is dominated by the Mediterranean and Indian Ocean.
- **Hybrid (machine-learning-corrected dynamical) approaches** — a neural correction applied to a subseasonal-to-seasonal (S2S) dynamical forecast (Sun et al. 2024), or a statistical SST model forced by a numerical-weather ensemble (Ross & Stock 2022) — are most advantageous at subseasonal lead times, at which purely data-driven models tend to lose spatial structure beyond approximately ten days.
- **Linear inverse models** (Wang et al. 2023) provide a parsimonious representation of MHW predictability and an interpretable connection to autoregressive state-space representations widely used in econometrics.
- **Statistical baselines** — climatology, persistence, damped persistence, and persistence-transformation preprocessing (Lee & Tzeng 2012) — together with analogue methods (Walsh et al. 2021, for air temperature in Alaska rather than SST) constitute the benchmark set against which the foregoing must be judged.

Illustrative results include dynamical SST anomaly correlations exceeding 0.8 in the tropics and far northern Pacific at the initial month (Smith & Spillman 2024) and day-one root-mean-square errors of 0.11–0.19 °C across Mediterranean sub-basins for deep-learning models (Bonino et al. 2024). Calibrated probabilistic verification — Brier skill scores, the symmetric extremal dependence index (SEDI), relative operating characteristic (ROC) analysis, and reliability diagrams — is now standard practice (Jacox et al. 2022; Liu et al. 2025; Smith & Spillman 2024).

3. Evidence on forecast skill

The nearest well-studied analogue to a Gulf-of-Alaska target is the northeast Pacific "Blob" region: Tang et al. (2024) employ NUIST-CFS1.0 with ensemble-Kalman data assimilation; Liu et al. (2025) apply SEAS5 across the northeast Pacific, including a Gulf-of-Alaska sub-region; and Taylor & Feng (2022) reproduce the 2014–2015 and 2019 events with a deep-learning model. These results pertain to the open northeast Pacific and, at most, the deeper Gulf of Alaska; none addresses the shallow, seasonally ice-covered Bering, Chukchi, or Beaufort shelves, where bottom thermal conditions and sea ice define the relevant predictand.

The Alaska shelf itself remains substantially unstudied. A structured review of a large high-latitude literature corpus (approximately 2,255 records) identified abundant fisheries, ecology, sea-ice, and impacts research but few studies reporting out-of-sample SST or MHW forecast skill relative to a benchmark. The recent review of machine-learning methods for MHW prediction by Welandawe et al. (2025) corroborates this independently: none of the studies it surveys falls within the sub-Arctic Alaska, Bering, or Barents domain. The direct high-latitude forecasting evidence is limited to a small number of studies, two of which are conference abstracts without quantitative verification (Langehaug et al. 2024 for the Barents Sea; de Boissésou et al. 2022 for a northeast Pacific precursor), together with Walsh et al. (2021), which addresses air temperature rather than SST.

4. Why high-latitude predictability might exist

In contrast to the paucity of methodological studies, the mechanisms underlying predictability in this region are comparatively well characterized, through two partly competing accounts.

The first is **atmospheric teleconnection sequencing**. The 2013–2014 onset of the northeast Pacific anomaly is attributed to anomalous atmospheric ridging and reduced ocean heat loss rather than to El Niño (Bond et al. 2015). Di Lorenzo & Mantua (2016) attribute the multi-year persistence of the 2014–2015 event not to static ocean memory but to a sequence of atmospheric forcing: an NPGO-like North Pacific pattern in 2014, linked through an extratropical–tropical teleconnection to a weak El Niño and returning as a PDO-like pattern in 2015, with modulation of the Aleutian Low forming part of the sequence. On this account, predictability resides in the atmosphere and in ENSO.

The second is **subsurface ocean memory and re-emergence**. Scannell et al. (2020) document subsurface anomalies preceding the 2019 event from 2017, insulated from the surface by a shallow mixed layer; Xu et al. (2026) relate ocean heat content to MHW persistence through seasonal re-emergence; and Amaya et al. (2021) provide the mechanistic basis, whereby a shoaling mixed layer amplifies the SST response per unit heat content ($\partial T/\partial t \approx Q/\rho c_p h$), accompanied by observed mixed-layer shoaling in the northeast Pacific and a record minimum in 2019. These accounts are not fully reconciled, and the forecasting evidence bears directly on the question: Tang et al. (2024) show that subsurface data assimilation is required to capture a Blob-scale event and that surface nudging alone is insufficient, indicating that the subsurface-memory channel should be represented in any operational system for the Gulf of Alaska.

The status of large-scale atmospheric precursors is more provisional. The Pacific Decadal Oscillation is well established as a pattern of persistence and forcing, but the Arctic Oscillation has not been established as a usable multi-year SST precursor: in the studies surveyed here it appears, at most, as a candidate atmospheric covariate, and the Aleutian Low appears principally within the teleconnection mechanism of Di Lorenzo & Mantua (2016) rather than as a predictor in its own right. Whether an upstream Arctic Oscillation–Aleutian Low–PDO chain confers genuine multi-year predictive skill for Gulf-of-Alaska SST, beyond the contributions of ENSO and of the subsurface ocean state, is an open research question rather than a settled result.

5. Outstanding methodological challenges

Four issues separate the validated literature from a credible Alaska shelf forecast.

First, **the Alaska shelf seas are not one system**. The Gulf of Alaska is deeper and more open-ocean-influenced, so that surface SST and MHW occurrence are the natural predictands; the Bering, Chukchi, and Beaufort shelves are shallow, strongly seasonal, ice-affected, and bottom-condition dependent, so that bottom thermal state and ice-conditioned occurrence matter more. Treating these as one "high-latitude North Pacific" problem obscures the fact that the relevant predictand, predictors, and benchmarks differ between them.

Second, **trend and forecastable interannual variability must be separated**. The distinction is sharp and consequential: undetrended skill may partly reflect secular warming, whereas detrended skill measures interannual predictive ability; both are useful, but they answer different questions. Zhang et al. (2023) provide the clearest illustration — in their system trend-only skill extends to approximately nine months, whereas the detrended interannual signal in the basin mean is significant only to approximately three months. Treatment of the trend is nonetheless inconsistent across the literature; some studies detrend, some report both treatments, and at least one (Smith & Spillman 2024) does not detrend and notes the resulting inflation of skill estimates. In a warming sub-Arctic record, conflating the two overstates genuine forecast skill.

Third, **the treatment of sea ice is unresolved**. Masking of ice-affected grid cells is not addressed in the assembled forecasting literature and is the least-developed requirement for any application to the Bering, Chukchi, or Beaufort shelves.

Fourth, **robustness across regime shifts is unestablished** for sub-Arctic shelves; Welandawe et al. (2025) identify rare-event and post-regime-shift performance as an unresolved, field-wide limitation.

6. Research-design implications

These limitations motivate a structured evaluation framework. A credible evaluation of MHW forecasting for Alaska shelf seas should compare a common set of methods — climatology, persistence, damped persistence, autoregressive and vector-autoregressive models, linear inverse models, dynamical seasonal forecasts, and hybrid machine-learning corrections — over common hindcast periods, under identical event definitions (a fixed MHW threshold and baseline

climatology), and with probabilistic verification (the Brier skill score and its decomposition, SEDI, ROC, and reliability diagrams). Trend should be handled explicitly and reported both ways, distinguishing undetrended skill from the detrended interannual component. The Gulf of Alaska and the Bering, Chukchi, and Beaufort shelves should be evaluated as separate systems, with predictands appropriate to each — surface SST and MHW occurrence for the former, bottom thermal state and ice-conditioned occurrence for the latter. Such a design would convert the present descriptive literature into a benchmarked, regionally specific evidence base. The central scientific problem is not the absence of forecasting methods; it is the absence of validated, benchmarked, trend-aware, probabilistic forecast evaluation for Alaska shelf systems.

Basis and limitations

This is a focused synthesis rather than a systematic review, intended to characterize the current state of the field and the evaluation problem it poses. It draws from a set of contemporary SST and MHW forecasting studies and a peer-reviewed review of machine-learning methods, together with a targeted reading of the North Pacific marine-heatwave mechanism literature. A small number of supporting points (for example, Scannell et al. 2020; Xu et al. 2026; the two conference abstracts) rest on abstract-level rather than full-text reading and are identified accordingly in the text. The principal limitation warrants emphasis: there is at present essentially no direct evidence on SST or MHW forecast methods for the Gulf-of-Alaska or Bering shelf, so every methodological recommendation here is an argument from transfer, and transfer to a shallow, strongly seasonal, ice-affected, and regime-shifting shelf remains an open question.

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